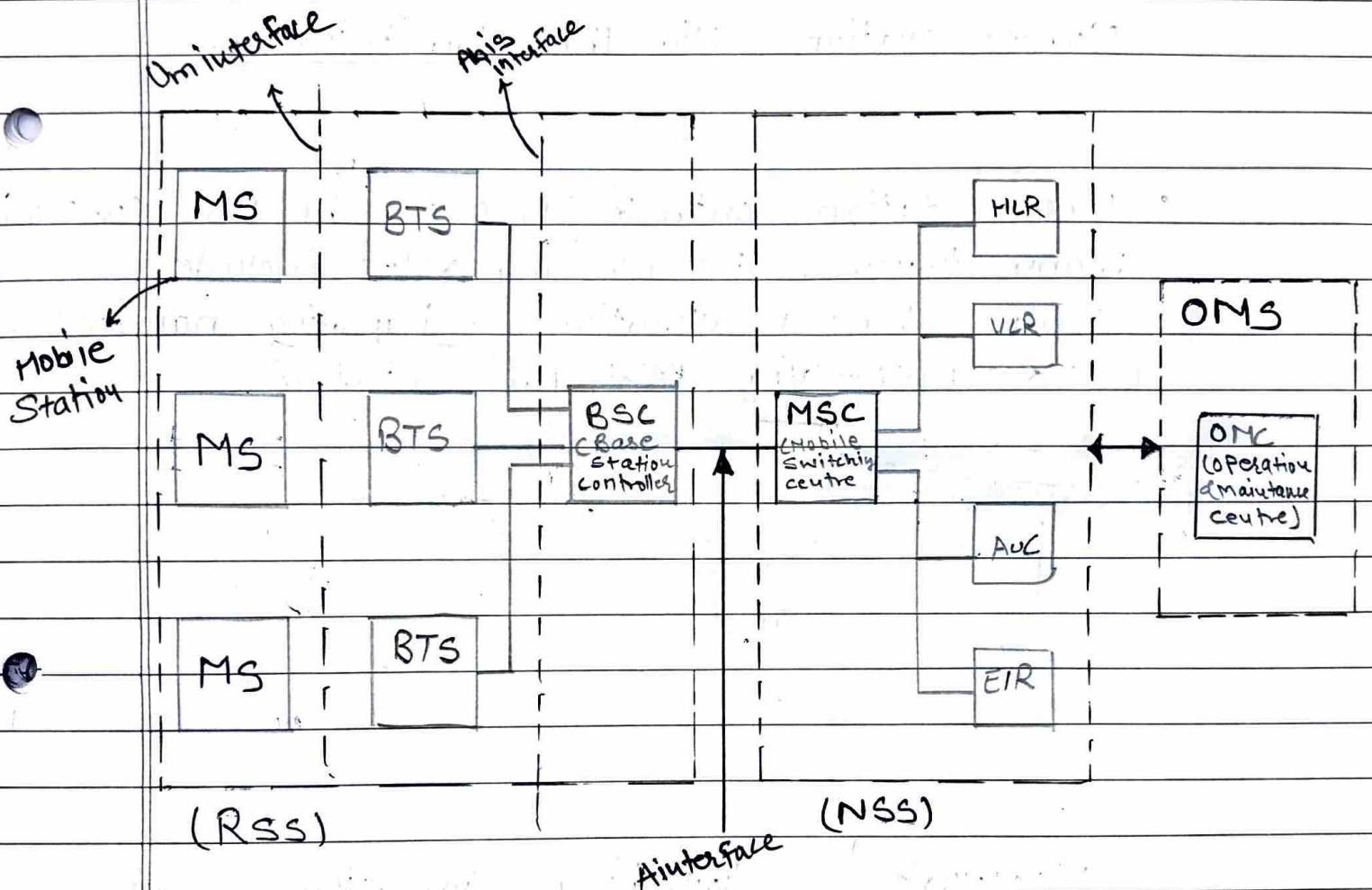


Q.1 GSM Architecture

The GSM architecture is a hierarchical system divided into three main subsystems: Radio Subsystem (RSS), Network Switching Subsystem (NSS), Operation & maintenance System (OMS).



1. Radio Subsystem (RSS)

The RSS handle Radio-specific tasks & connect the mobile device to the core network.

- Mobile station (MS): This mobile station is our

phone and Sim card. The Sim card plays important role because it stores the identity of a user.

- Base Transceiver Station (BTS) :- Base transceiver is a radio tower that defines a cell. The BTS connects & communicates with Mobile station via the Um interface.

- Base Station Controller (BSC) :- The BSC controls many towers. Its primary roles include radio channel allocation & handling handoffs. $BSC \leftrightarrow MSC$ via ~~BSS~~ Abis interface

(Simple) :- Ham travel krte hai or usi se phone pe baat kr rha hai or apna network Tower A se connect kr rha hai aur jab ham travel krte hai dur jate rehte hai & Tower A ka connection weak hota hai or wo automatically Tower B se connect hojata without any end call)

2. Network & Switching Subsystem :- This part handles the calling logic & keeps all your data safe.

- Mobile Switching Centre :- It is the boss of network. It connects all our calls, setup & connection to other network like PSTN.

$MSC \leftrightarrow BSC$

Via Abis interface

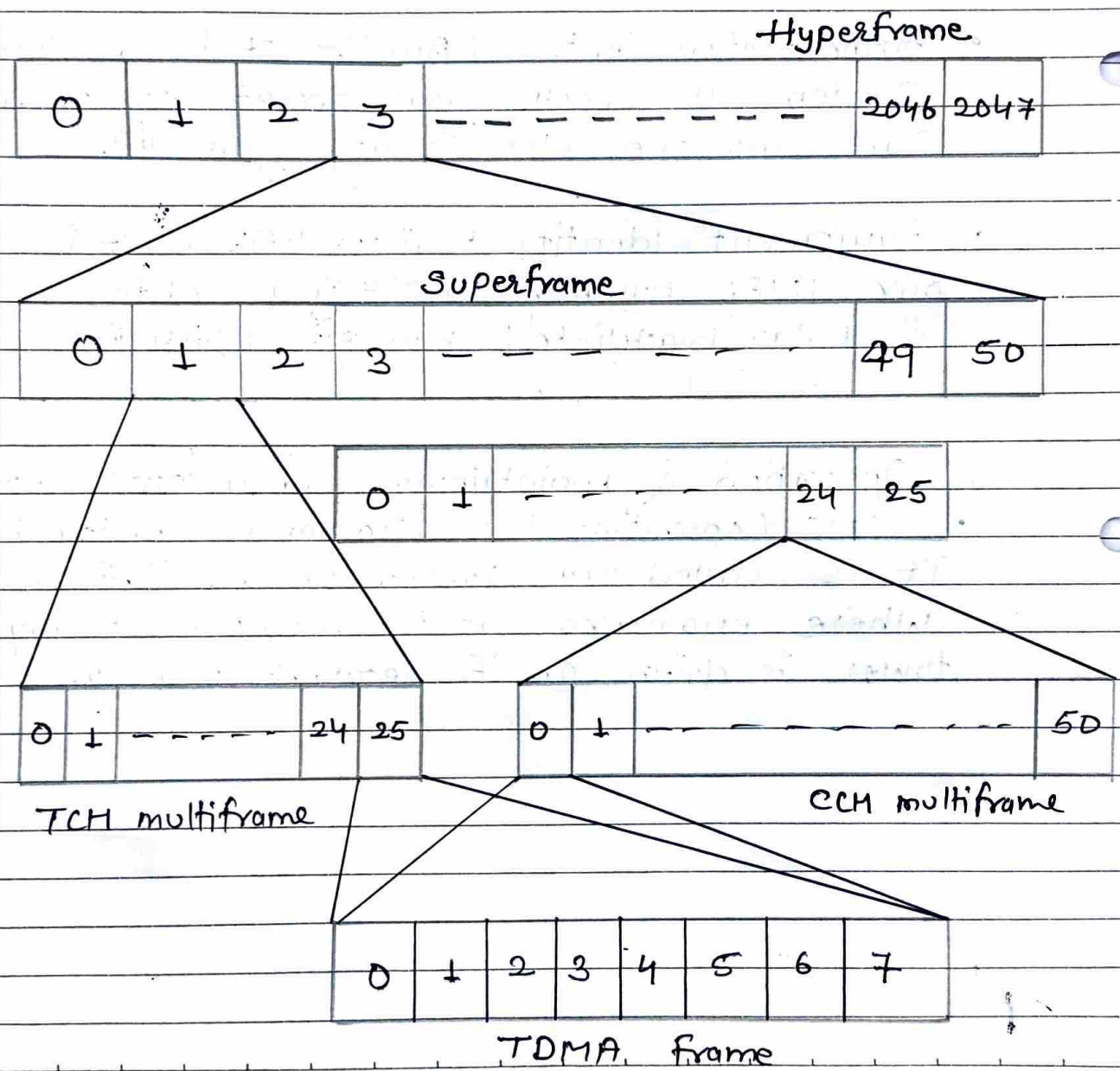
- Home Location Register (HLR):- It is a permanent database containing sub subscriber profiles, service details & current location information.
- Visitor location Register (VLR):- It is a temporary database that store data for subscriber currently roaming in that specific area.
- Authentication Centre (AuC):- It is a Secure System. It checks your secret key to make sure no one else using your SIM.
- Equipment Identity Register (EIR):- It is checks out IMEI numbers. If its your phone is stolen & later blacklisted then EIR block it.

3. Operation & maintaining Subsystem (OMS)

- OMC (Operation & maintenance Centre):- It is called as Control room. This is where engineers sit to monitor if any tower is down or if network is busy.

Q. GSM time slot hierarchy.

GSM uses TDMA (Time Division Multiple Access) to allow multiple user to share the same frequency. To manage this data is stored into a strict hierarchy starting from a single Time slot up to a Hyperframe.



1. Time slot (Smallest Unit)

- This is the basic unit of data
- Duration: 0.577 ms
- It contains the burst of data (156.25 bits)

2. TDMA frame

- It contains 8 Time slot (0 to 7) combine to form 1 TDMA frame
- Duration: 4.615 ms
- Each of 8 slot is used by a different user.

3. Multiframe

a. TCH :- Contains 26 TDMA frames. It carries voice/speech.

- Duration: 120 ms

b. CCH :- Contains 51 TDMA frame. It contains network signal

- Duration: 235.4 ms

4. Superframe:- This is larger group

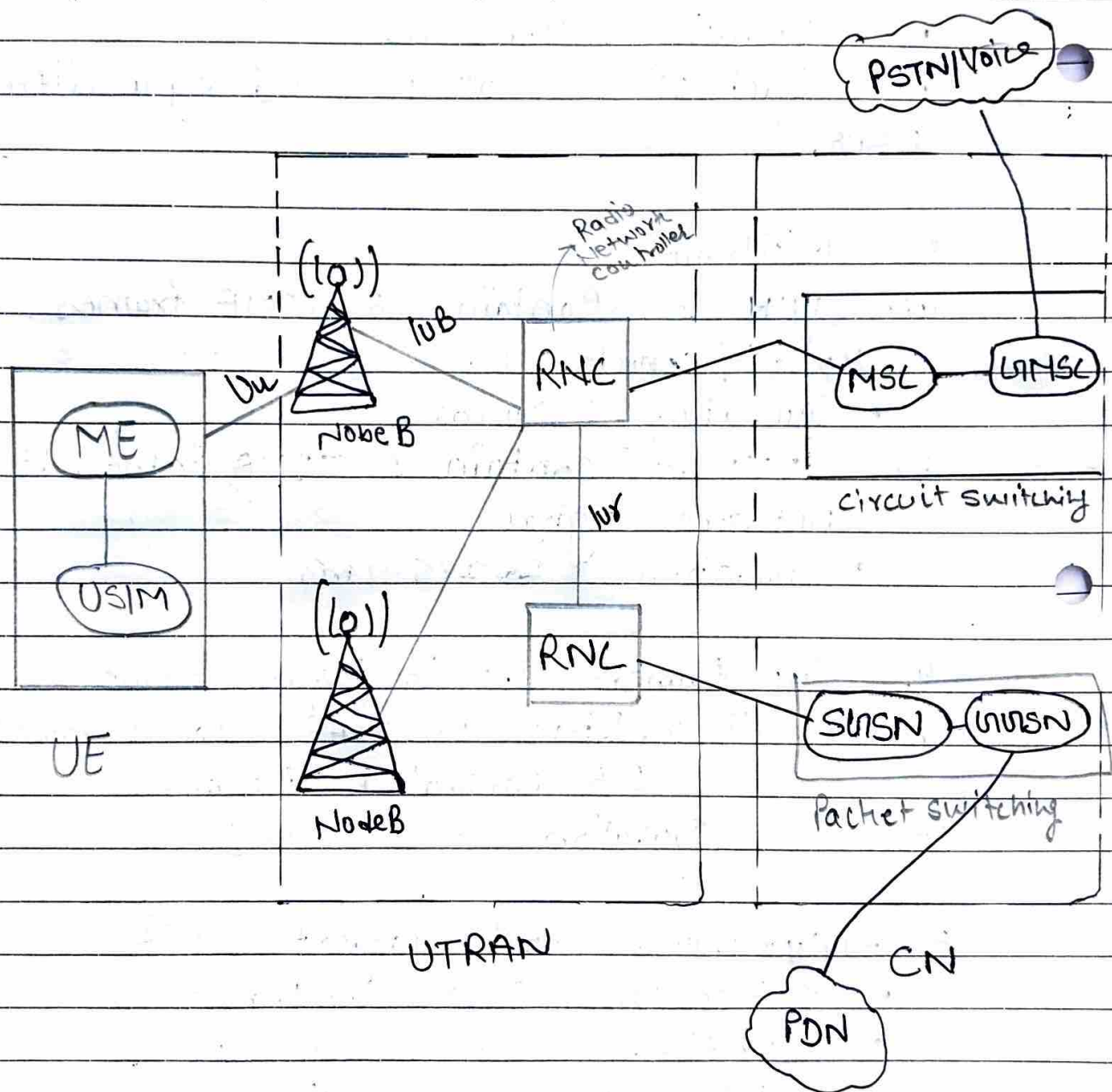
- It consists 51 traffic multiframe & 26 control multiframe
- Duration: 6.12 second

5. Hyperframe:- It is largest Unit

- It contains 2048 superframes
- Duration: 3 hour 28 minutes
- It is used in encryption so no-one can hack or listen your call.

Q. Draw & explain UMTS (3G) Architecture.

UMTS stands for Universal Mobile Telecommunication System. UMTS is a 3G mobile technology that provides high-speed data & voice services. UMTS architecture divide into three main domains: User Equipment (UE), UTRAN & Core Network (CN).



1. User Equipment (UE)

It is simply the device in your hand

- ME (mobile Equipment) :- The physical 3G handset
- USIM (Universal Subscriber Identity Mobile) :- It is an advanced SIM card that provides better security & support 3G services.

2. UTRAN^A :- This is the radio network that handle wireless connection.

- NodeB :- It is the just 3G tower. It talks the network of your phone via Uu interface
- RNC (Radio Network Controller) :- This is the brain of towers. It decides which tower you should be connected & manages the speed of your data.

Interfaces :-

Uu :- Between UE & Node B

Iub :- Between Node B & RNC

Iur :- Between two different RNC

3. Core Network (CN) :- The CN is responsible for switching calls & routing data. It's divided into two parts:

① Circuit Switched (CS) domain :- It's handle voice calls.

- It consist MSC & GMSC which connect PSTN

② Packet Switched (PS) domain :- It's handle data internet.

- It consist SGSN & GGSN which connect to PDN.

Q. GSM VS UMTS

GSM (2G)

- ① Global System for Mobile Communication
- ② 2nd generation
- ③ technology based on TDMA & FDMA
- ④ Slow (9.6Kbps)
- ⑤ It's tower called BTS
- ⑥ BSC control GSM
- ⑦ Sim Card used standard
- ⑧ Core Network are HLR, VLR, AUC, EIR

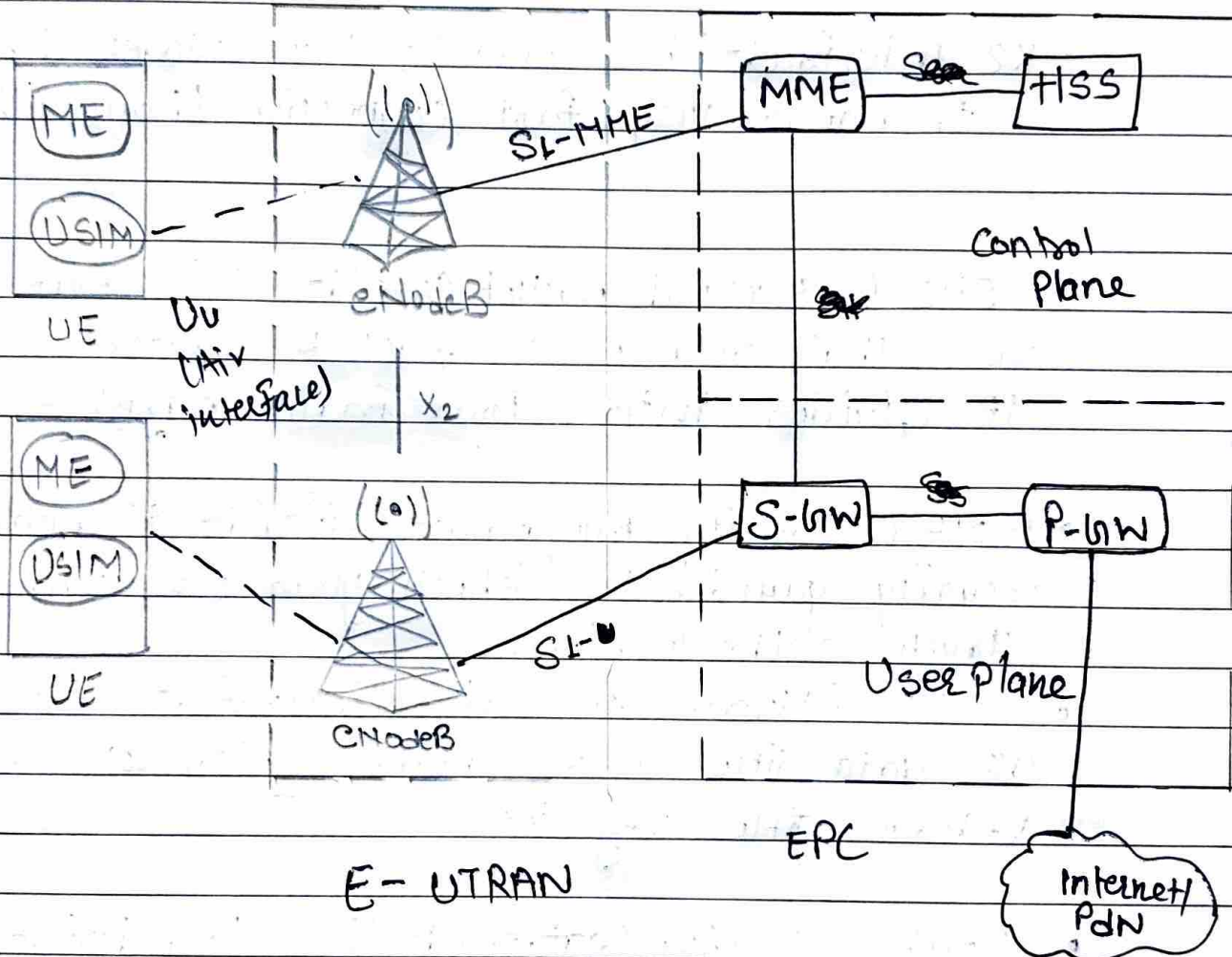
UMTS (3G)

- ① Universal mobile telecommunications System
- ② Third generation (3G)
- ③ W-CDMA (wideband-CDMA)
- ④ Fast (2Mbps)
- ⑤ Node B
- ⑥ RNC
- ⑦ USIM (Universal SIM)
- ⑧ SGSN, SSAN & SSAN
GGSN

Bati piche dono he
bare me padha hai
likh dena!!

8. LTE (4G) Architecture

Long-term Evolution architecture is designed to provide high-speed data with low latency. It is divided into three main parts.



1. UE

ME → handset 4G

USIM → SIM Card → Security

2. E-UTRAN :- It is radio network. In 4G there is only one entity here which make it faster.

- eNodeB :- It is a smart tower. It sends the signal and also decide the speed of signal it communicate with User equipment via Air interface.

- X2 Interface :- This connects two eNodeB directly to handle very fast connections/handovers

3. EPC (Evolved packet Core) :- It is flat architecture that only uses packet switching. It divides into four main parts.

- MME (Mobility management Entity) :- It's behave like security guard. It's check your password & track the location.

- HSS (Home Subscriber Server) :- It is maintain all data like phone number, plan, profile etc. Behave like register.

- S-GW (Serving Gateway) :- It's data bridge that routes data between eNodeB & core network.

- P-GW (PDN Gateway) :- Give ip & connect to server like google, youtube etc.

4. Interfaces

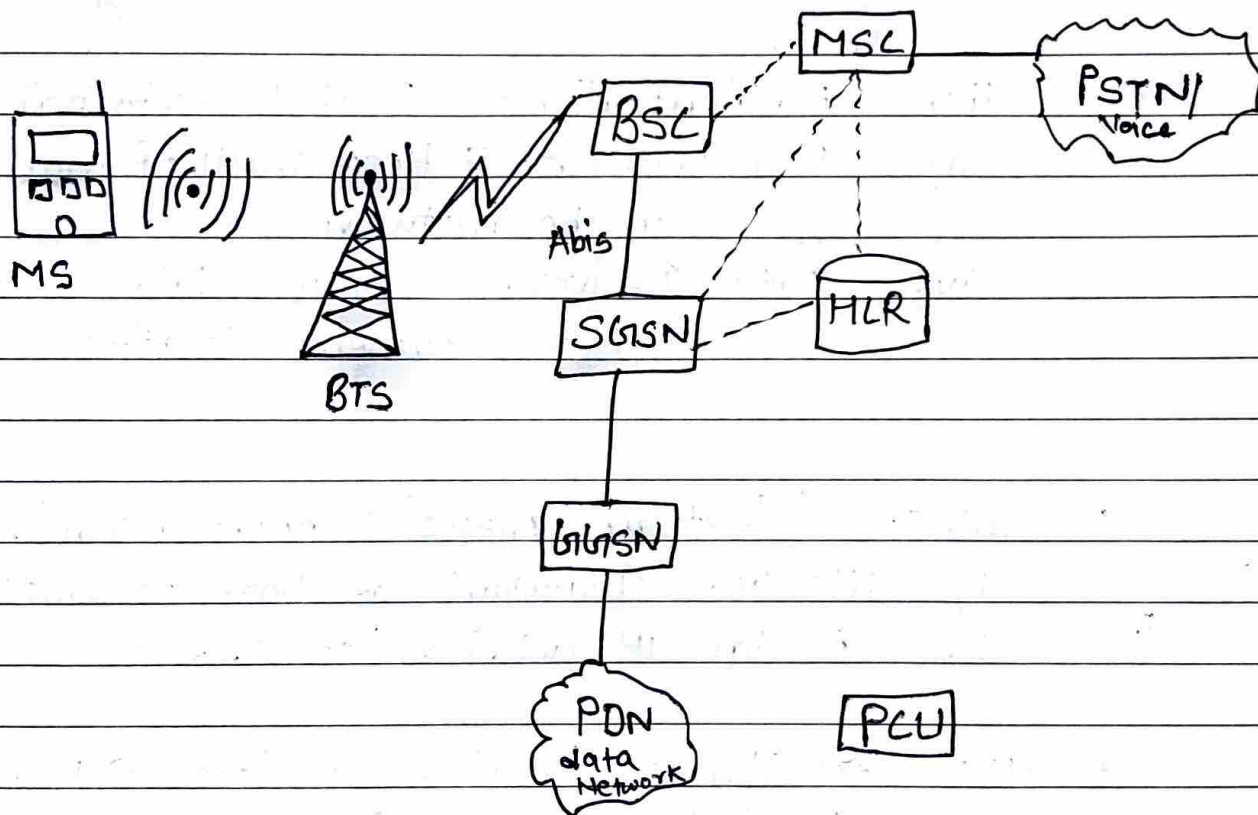
U₀ → Phone to tower

X₂ → b/w eNodeB, S₁ → b/w tower & core network.

Q. Explain GPRS Architecture and the roles of SGSN & GGSN.

GPRS (General packet radio Service) is a technology that allows 2G mobile phones to send and receive data over the internet. It's add packet-switched path to old GSM network.

GPRS has two main component: SGSN & GGSN



Features are in GPRS

- Mobile station :- This is user's mobile phone. It connects to the network via BTS (tower)

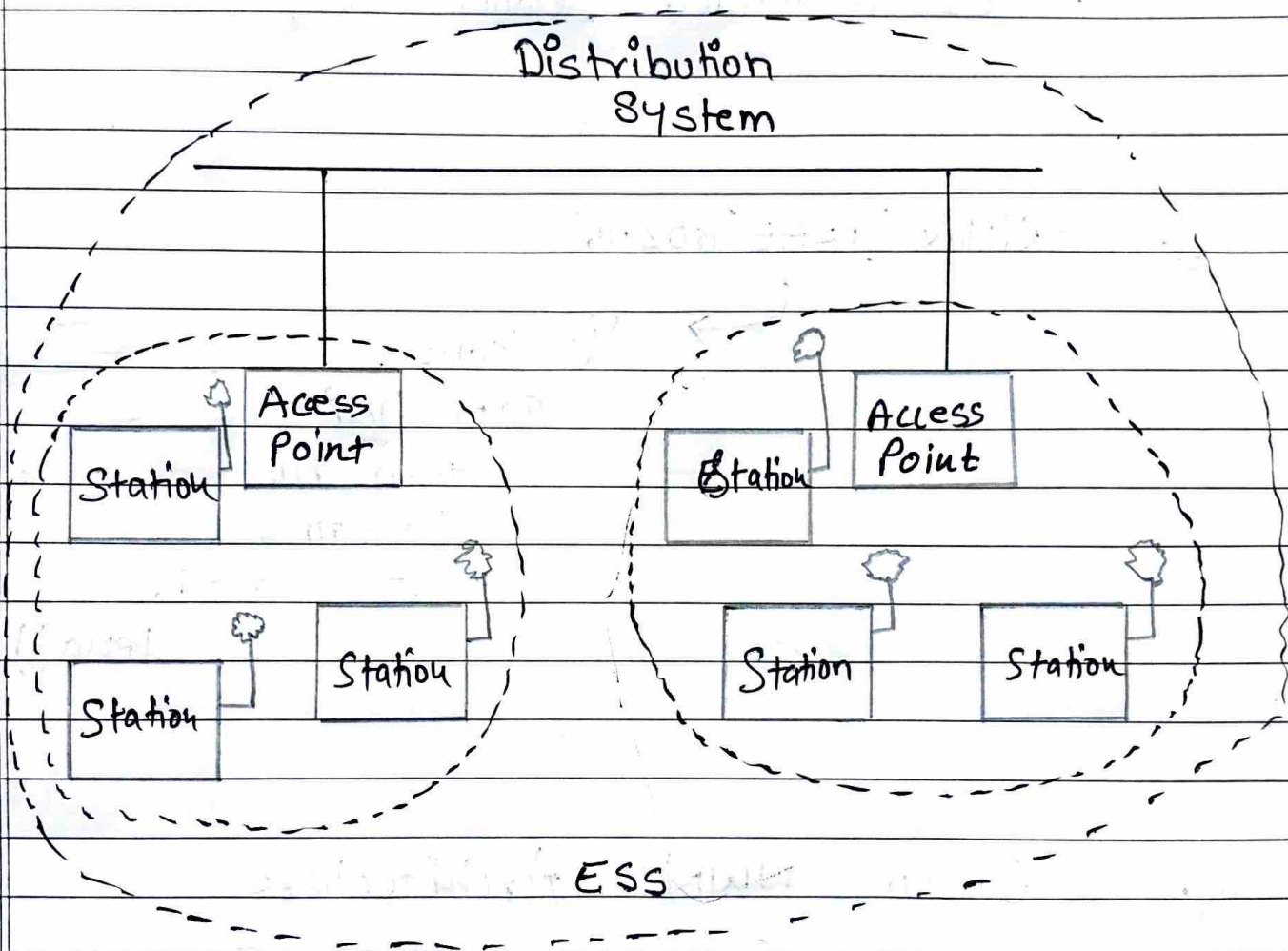
↳ Base transceiver station

- BSC (Base Station Controller):- It manages the Radio signals from the Base transceiver receiver (BTS)
- PCU (Packet Control Unit):- It is the new part of BSC to improve voice calls from internet data
- SGSN (Serving GPRS Support Node)
 - i) It acts like "a local manager" for user.
 - ii) Authentication:- ID check karta hai User ki.
 - iii) Registration:- It keep track of mobile device in the network
 - iv) Data Delivery:- routes data packets at user's current location.
- GGSN (Gateway GPRS Support Node):-
 - i) It's like 'Gateway' or 'Door' to Outside world
 - ii) Assign IP address to the user
- HLR → (Home location Register) → Permanent User information Store

Simple:- SGSN ka kaam ye hai ki wo dekhta hai ki hum kaha or Sahi user ho ya Nahi (Security)
 GGSN ka kaam humhe Internet se Jodna (Youtube)
 PCU decide karta hai Voice call hai ya Internet calling data.

Q. Draw & explain system architecture of IEEE 802.11.
11. Difference b/w IEEE 802.11 & IEEE 802.16.

Ans. IEEE 802.11 is a WiFi architecture that consists of several components that work together to provide wireless connectivity.



Components:-

- Station (STA):- The most basic Component of the network. Any device that contain an IEEE 802.11 compliant MAC & physical layer interface.
eg; Laptop, Smartphone.

• AP:- A station that provide access to the distribution service via wireles medium.

• BSS —

• DS (Distribution System)

• ESS (Extended Service Set)

Self
explain
xyo

for better
understanding

Q. WiMax (IEEE 802.16)



Ye kam

aata hai

kyun ho

toh

padh

lena!!

Q. Cisco UWN Architecture



Exam se phle

hinge ye!!

Q1. Differentiate between FDMA, TDMA, and CDMA. (10 Marks)

Feature	FDMA	TDMA	CDMA
Full Form	Frequency Division Multiple Access.	Time Division Multiple Access.	Code Division Multiple Access.
Basic Concept	Bandwidth is divided into small frequency channels.	Entire bandwidth is used in different time slots.	All users use the full bandwidth at the same time.
Separation	Users are separated by Frequency.	Users are separated by Time.	Users are separated by Unique Codes.
Efficiency	Low (Bandwidth is wasted in empty channels).	Medium (Users must wait for their turn/slot).	High (Bandwidth is used 100% all the time).
Handover	Only Hard Handover is possible.	Hard Handover is used.	Soft Handover (Very smooth, no call drops).
Complexity	Simple hardware design.	Medium (Needs strict clock synchronization).	Very high (Needs complex mathematical coding).
Security	Low (Easy to intercept signals).	Medium security.	Very High (Impossible to decode without the code).
Network	Used in 1G (Analog).	Used in 2G (GSM).	Used in 3G (UMTS).

SIMPLE:- **FDMA** matlab har user ko alag frequency dena (jaise alag road). **TDMA** matlab sab ek hi road use karenge par baari-baari (jaise traffic signal). Aur **CDMA** matlab sab ek saath ek hi road par hain par sabki gaadi ki language (code) alag hai, toh koi kisi se takrayega nahi.

Q2. Explain Spread Spectrum and Differentiate between FHSS and DSSS. (10 Marks)

Spread Spectrum is a technique where the transmitted signal occupies a bandwidth much wider than the minimum necessary to send the data. It spreads the power of the signal over a wide range to make it secure.

1. Frequency Hopping Spread Spectrum (FHSS):

- **How it works:** The carrier frequency "hops" across different channels in a pseudo-random sequence.
- **Hopping Rate:** It can be Fast (multiple hops per bit) or Slow (multiple bits per hop).
- **Resistance:** If one frequency is jammed, the signal moves to another, making it very hard to block.

2. Direct Sequence Spread Spectrum (DSSS):

- **How it works:** Each bit of data is multiplied by a high-speed "Chipping Code" (Pseudo-noise code). This spreads the signal across the entire band.
- **Signal Appearance:** To an outsider, the signal looks like low-level noise.
- **Advantage:** It provides very high data security and is great for multi-path environments.

Feature	FHSS	DSSS
Full Form	Frequency Hopping Spread Spectrum	Direct Sequence Spread Spectrum
Mechanism	Carrier hops across many frequencies.	Signal is spread using a Chipping Code.
Bandwidth Use	Uses a small part of bandwidth at a time.	Uses the full wide band all the time.
Security	High (Hard to follow the hop).	Very High (Signal looks like noise).
Hardware	Simple and cheap.	Expensive and complex.
Speed	Low data rates.	High data rates.

Resistance	Resistant to "Narrow-band" noise.	Resistant to "Multi-path" fading.
Examples	Used in Bluetooth.	Used in Wi-Fi and GPS.

SIMPLE: - **Spread** Spectrum ek technique hai jisme signal ko wide bandwidth me spread kiya jata hai taaki communication secure rahe aur noise ya jamming ka effect kam ho. Iske do main types hote hain: FHSS aur DSSS. FHSS (Frequency Hopping Spread Spectrum) me signal continuously different frequencies par hop karta rehta hai, isliye agar ek frequency jam ho jaye toh communication dusri frequency par continue ho jata hai. Ye Bluetooth me use hota hai. DSSS (Direct Sequence Spread Spectrum) me data ko chipping code ke saath spread kiya jata hai, jisse signal outsider ko noise jaisa dikhta hai aur security zyada milti hai. Ye Wi-Fi aur GPS me use hota hai. FHSS simple aur low speed hota hai, jabki DSSS high speed aur more secure hota hai.

Q3. Describe the Evolution of Mobile Communication from 1G to 5G. (10 Marks)

1. 1G (First Generation - 1980s):

- **Technology:** Analog Signals (AMPS).
- **Capability:** Voice only.
- **Limitations:** Poor voice quality, no security, and huge phones.

2. 2G (Second Generation - 1990s):

- **Technology:** Digital Signals (GSM/CDMA).
- **Capability:** Voice + SMS (Text messages).
- **Key Intro:** SIM cards and Encryption were introduced.

3. 3G (Third Generation - 2000s):

- **Technology:** W-CDMA (UMTS).
- **Capability:** High-speed internet + Video Calling.
- **Key Intro:** The ability to browse the web and use basic apps on phones.

4. 4G (Fourth Generation - 2010s):

- **Technology:** LTE (Long Term Evolution).
- **Capability:** HD Video Streaming, Gaming, and High-Speed Data.
- **Key Intro:** Purely IP-based network (VoLTE). No more traditional switching.

5. 5G (Fifth Generation - 2020s):

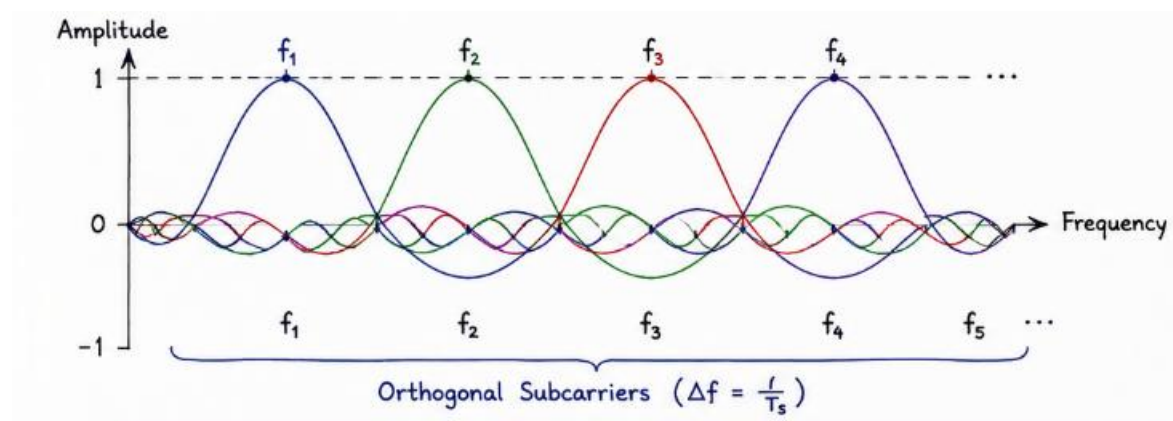
- **Technology:** mmWave and Massive MIMO.
- **Capability:** 10 Gbps speeds, zero lag (latency), and IoT.
- **Key Intro:** Connects millions of devices like self-driving cars and smart homes.

Q4. Explain OFDMA Technique for Multiplexing with a diagram. (10 Marks)

Answer:

1. Introduction:

- **OFDMA** stands for **Orthogonal Frequency Division Multiple Access**.
- It is a multiplexing technique used in **4G LTE** and **5G** networks.
- It allows multiple users to share the same frequency band at the same time.



- **Sub-carriers:** In OFDMA, the total available bandwidth is divided into many small frequency bands called **Sub-carriers**.

- **Orthogonality:** The key feature, as seen in the diagram, is that these sub-carriers are **Orthogonal**. This means they are placed in such a way that they can overlap without interfering with each other.
- **Efficient Spectrum Use:** Because the sub-carriers can overlap, there is no need for **Guard Bands** (empty spaces) between frequencies. This uses the available spectrum very efficiently.
- **Multi-user Access:** The sub-carriers are grouped and dynamically assigned to different users based on their data needs at that specific time.

3. Key Advantages:

1. Provides **high data speeds**.
2. Supports a **massive number of users** simultaneously.
3. Highly resistant to signal fading caused by interference.

simple: - OFDMA mein ek bada internet "road" (bandwidth) hai, jise humne thousands of choti-choti "lanes" (sub-carriers) mein baant diya hai. Sabse bada fayda ye hai ki ye lanes ek-dusre ke upar "overlap" ho sakti hain (isi ko Orthogonal kehte hain) bina accident hue, isliye road ki koi jagah waste nahi hoti. Alag-alag users ko unki zarurat ke hisaab se lanes mil jati hain, jiske wajah se 4G aur 5G fast chalta hai.

Q5. Differentiate between Infrastructure WLAN and Ad-hoc WLAN. (10 Marks)

Answer:

Wireless Local Area Networks (WLAN) can operate in two primary modes depending on how the devices are connected.

Feature	Infrastructure WLAN	Ad-Hoc WLAN (IBSS)
Topology	It is a centralized network.	It is a decentralized (Peer-to-Peer) network.
Central Device	Requires an Access Point (AP) or Wireless Router.	No central device is required.
Connectivity	Devices communicate through the AP.	Devices communicate directly with each other.
Scalability	Easy to add more devices and extend the range.	Difficult to manage as the number of devices increases.
Range	Range is defined by the Access Point's coverage.	Limited to the radio range of the individual devices.
Complexity	More complex to set up initially.	Very simple and quick to set up.
Cost	Higher cost due to the need for an AP/Router.	Low cost as no extra hardware is needed.
Stability	More stable and better for permanent setups.	Less stable; if one device moves away, the link breaks.

Simple :- Infrastructure WLAN matlab ek system jisme ek "Boss" (Access Point/Router) hota hai. Sab devices pehle usse connect hote hain, tabhi data transfer hota hai. Jaise tumhare college ka Wi-Fi.

Ad-hoc WLAN matlab jab do devices aapas mein bina kisi router ke connect ho jayein. Jaise do laptops ke beech "Direct Wi-Fi" ya "Bluetooth" se file share karna. Ye temporary hota hai aur isme koi central router nahi chahiye hota.

Q6. Explain CSMA/CA and its Access Mechanisms (DCF and PCF). (10 Marks)

Answer:

1. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

Since wireless media cannot detect collisions while transmitting (unlike wired Ethernet), it uses **Collision Avoidance**.

- **Carrier Sense:** The device "listens" to the channel. If it is busy, it waits.
- **Collision Avoidance:** If the channel is free, the device waits for a small random time (Backoff) before sending data to prevent two devices from starting at the exact same moment.

2. DCF (Distributed Coordination Function)

DCF is the **mandatory** and most common mechanism. It is **contention-based**, meaning every device fights for the channel.

- **How it works:** It uses the **CSMA/CA** protocol.
- **IFS (Inter-Frame Spacing):** Devices wait for a period called **DIFS** (DCF Inter-Frame Spacing).
- **Binary Exponential Backoff:** If the channel is busy, the device picks a random wait time. This time doubles every time a collision occurs.
- **Best for:** Distributed, simple networks like home Wi-Fi.

3. PCF (Point Coordination Function)

PCF is an **optional** mechanism. It is **contention-free**, meaning there is no fighting for the channel.

- **How it works:** The **Access Point (AP)** acts as a "Point Coordinator." It performs **Polling**.
- **Polling:** The AP asks each device one by one: *"Do you have data to send?"*
- **Priority:** PCF has higher priority than DCF. It uses a shorter wait time called **PIFS** (PCF Inter-Frame Spacing).
- **Best for:** Time-sensitive data like Voice or Video (Real-time traffic).

CSMA/CA ka matlab hai "pehle suno, phir thoda ruko, phir bhejo" taaki signals aapas mein na takrayein.

1. **DCF** matlab "Azaad system." Har mobile khud decide karta hai kab data bhejna hai. Agar line busy hai, toh wo random time wait karta hai.
2. **PCF** matlab "Monitor wala system." Yaha Router (AP) ek monitor ki tarah sabse baari-baari puchta hai ki "Kya tumhe data bhejna hai?". Isme koi ladayi nahi hoti kyunki Router sabko line se chance deta hai.

Q7. Differentiate between MANET, WSN, and VANET. (10 Marks)

Answer:

These are all types of wireless networks, but they are used for very different purposes and have different technical characteristics.

Feature	MANET	WSN	VANET
Full Form	Mobile Ad-hoc Network.	Wireless Sensor Network.	Vehicular Ad-hoc Network.
Primary Nodes	Laptops, Mobile phones.	Sensor nodes (small).	Vehicles (Cars, Trucks).
Mobility	Low to Medium.	Usually Static (Fixed).	Very High (High speed).
Power Source	Replaceable batteries.	Limited battery (Hard to replace).	Continuous power from engine.
Topology	Changes frequently.	Mostly constant.	Changes very rapidly.
Purpose	General data sharing.	Monitoring environment (Temp, light).	Traffic safety and navigation.
Reliability	Medium.	High (due to many nodes).	High (for safety/life).
Range	Moderate.	Short (Node-to-node).	Long (High power radio).

Q8. Compare WEP and WPA2 Wireless Security Protocols. (10 Marks)

Answer:

WEP and WPA2 are security protocols used to protect Wi-Fi networks by encrypting data. Over time, WEP has become obsolete due to security flaws, while WPA2 is the current standard for most networks.

Feature	WEP	WPA2
Full Form	Wired Equivalent Privacy.	Wi-Fi Protected Access 2.
Encryption Algorithm	Uses RC4 (Stream Cipher).	Uses AES (Advanced Encryption Standard).
Key Length	40-bit or 104-bit (Short).	128-bit to 256-bit (Long/Strong).
Key Management	Static key (Doesn't change).	Dynamic keys using CCMP.
Security Level	Very Weak (Easy to crack).	High (Very secure).
Integrity Check	Uses CRC-32 (Weak).	Uses CBC-MAC (Strong).
Authentication	Shared Key or Open system.	Uses PSK (Pre-Shared Key) or Enterprise.
Current Status	Deprecated/Obsolete.	Most widely used standard.

1. WEP (Wired Equivalent Privacy)

- **Mechanism:** It was the first security protocol for Wi-Fi. It uses a static key that must be manually entered into every device.
- **Weakness:** Because the key is static and short, hackers can easily collect enough data packets to "crack" the key in just a few minutes using simple software.

2. WPA2 (Wi-Fi Protected Access 2)

- **Mechanism:** It replaced WEP and WPA. It uses **AES** encryption, which is the same standard used by governments to protect top-secret data.
- **Security:** It uses a four-way handshake to verify the password without actually sending the password over the air. It also changes the encryption keys for every packet sent, making it extremely difficult to hack.

simple logic ye hai ki **WEP** purane zamane ka lock hai jo ek patli taar se bhi khul jata hai. Isme password change nahi hota tha aur hackers ise 2 minute mein tod dete the.

WPA2 aaj kal ka modern digital lock hai. Isme **AES** encryption use hota hai jo itna strong hai ki use todne mein saalon lag jayenge. Aaj hum jo bhi Wi-Fi password set karte hain, wo zyada tar WPA2 hi hota hai kyunki ye sabse safe hai.

Q9. Explain the security flaws associated with Initialization Vector (IV) in WEP. (5 Marks)

The Initialization Vector (IV) is a 24-bit field used in WEP to ensure that the same plaintext does not produce the same ciphertext. However, it has major design flaws:

1. **Small IV Size (24-bit):** The most critical flaw is that the IV is only 24 bits long. This provides only about 16 million possible values. On a high-speed busy network, these combinations are exhausted in a few hours, leading to **IV reuse**.
2. **IV Collision:** When the small pool of IVs is exhausted, the network starts repeating the same IVs. When a hacker captures two different packets using the same IV, they can use statistical analysis to recover the key stream and eventually the secret password.
3. **Clear Text Transmission:** WEP sends the IV in the header of each packet as **clear text** (unencrypted). This means any attacker sniffing the network can easily read the IV values without needing any special decryption keys.
4. **Weak Key Generation:** Certain IV values are considered "weak." When these weak IVs are combined with the secret key, they leak information about the password bits, allowing hackers to use mathematical attacks to crack the key.
5. **Lack of Key Rotation:** WEP does not have a mechanism to automatically change or rotate the secret keys. Combined with the short IV, this gives hackers unlimited time to collect enough packets to identify a pattern and break the encryption.

simple language mein WEP IV ke 5 bade masle (problems) ye hain:

1. **Chota Size:** IV sirf 24-bit ka hai, jo bahut jaldi khatam ho jata hai.
2. **Repetition:** Jab numbers khatam hote hain, toh wahi IV repeat hone lagta hai, jise "Collision" kehte hain. Hacker isi repetition ka fayda uthate hain.
3. **Khule aam bhejiyo:** IV ko packet ke upar bina kisi security ke "Clear Text" mein bheja jata hai, toh hacker ise aram se "sniff" (dekh) sakta hai.
4. **Kamzor Numbers:** Kuch IV numbers weak hote hain jo password guess karna aasaan bana dete hain.
5. **Fixed Password:** WEP mein password badalta nahi hai, isliye hackers ko pattern dhoondne ka poora time mil jata hai.

Q.10 Compare the security features of GSM (2G) and UMTS (3G).

GSM was designed for basic voice security, but it had several weaknesses. UMTS was developed to fix these flaws and provide much stronger security for data and video.

Feature	GSM Security (2G)	UMTS Security (3G)
Authentication	Unilateral: Only the network checks the user.	Mutual: User checks the network AND network checks the user.
Encryption Key	64-bit (Weak).	128-bit (Strong).
Integrity Check	No data integrity check.	Strong data integrity (prevents data tampering).
Algorithms	A3, A5/1, A8 (Many are now broken).	KASUMI (UEA1) and SNOW 3G (Stronger).
Identity Protection	Uses IMSI/TMSI (IMSI Catchers can hack it).	Better TMSI management (Harder to track).
Visibility	Security is invisible to the user.	User can see if encryption is ON or OFF.
SIM Card	Standard SIM.	USIM (Universal SIM) with higher security.
Encryption Scope	Only between Phone and Base Station.	From Phone up to the Core Network (RNC).

SIMPLE:-

1. **GSM** purana system tha jisme sirf network tumhara password check karta tha. Iska fayda utha kar hackers fake tower laga lete the aur tumhara phone bina puche connect ho jata tha. Iski encryption bhi kamzor thi.
2. **UMTS (3G)** ne isse theek kiya. Isme **Mutual Authentication** hota hai—matlab network tumhara password check karega aur tumhara phone bhi network ka ID check karega. Agar tower fake hua, toh tumhara phone connect nahi hoga. Isme encryption keys double (128-bit) hain, toh ise todna mushkil hai.

Q.11 Explain the Digital Signature process in detail.

A **Digital Signature** is a mathematical technique used to validate the authenticity and integrity of a digital document. It ensures that the message was created by a known sender and was not altered during transit.

The Digital Signature Process (Step-by-Step)

The process involves two main phases: **Signing** (by the sender) and **Verification** (by the receiver).

Phase 1: Signing Process (Sender's Side)

1. **Hashing:** The sender takes the original message and passes it through a **Hash Function** (like SHA-256) to create a fixed-size string called a **Message Digest** (Hash).
2. **Encryption:** The sender then encrypts this Message Digest using their own **Private Key**.
3. **Digital Signature:** This encrypted hash is the **Digital Signature**.
4. **Combining:** The sender attaches the Digital Signature to the original message and sends it to the receiver.

Phase 2: Verification Process (Receiver's Side)

1. **Separation:** The receiver receives the package and separates the original message from the Digital Signature.
2. **Decrypting Signature:** The receiver decrypts the Digital Signature using the **Sender's Public Key** to get the original Hash (Digest A).
3. **Hashing Original Message:** The receiver takes the received message and runs it through the same Hash Function to create a new Hash (Digest B).

4. **Comparison:** The receiver compares Digest A and Digest B.

- If **Digest A = Digest B**, the signature is valid (Integrity and Authenticity confirmed).
- If they do not match, the document has been tampered with.

SIMPLE: -Digital Signature ka matlab hai ek "Digital Mohar" (Stamp). Isme sender pehle message ka ek short code (Hash) banata hai aur use apni **Private Key** se lock kar deta hai. Receiver ke paas sender ki **Public Key** hoti hai jisse wo us lock ko kholta hai. Agar lock khul jaye aur message ka hash match ho jaye, toh iska matlab hai ki message raste mein kisi ne badla nahi hai aur ye sahi insaan ne hi bheja hai. Simple logic: Private Key se sign hota hai, aur Public Key se check hota hai.

Q12. Explain Mobility Support in Cisco Unified Wireless Network (UWN). (10 Marks)

Answer: Mobility in Cisco UWN allows a wireless user to move from one Access Point (AP) to another without losing their network connection (Roaming). Cisco manages this through **Mobility Groups** and **Controllers (WLC)**.

1. Mobility Architecture (How it Works)

Cisco UWN uses a centralized architecture where the **Wireless LAN Controller (WLC)** manages the mobility state.

- **Layer 2 Roaming:** The client moves to a new AP but stays within the same subnet. The WLC simply updates the database with the new AP location.
- **Layer 3 Roaming:** The client moves to a new AP that belongs to a **different subnet**. To keep the connection alive, Cisco uses a "Tunneling" mechanism:
 - **Anchor Controller:** The original WLC where the user first connected.
 - **Foreign Controller:** The new WLC where the user has moved.
 - **Tunnel:** Data is sent from the Foreign Controller back to the Anchor Controller so the user's IP address doesn't have to change.

2. Mobility Messaging Process

1. **Handshake:** When a client roams, the new WLC sends a "Mobility Announcement" to other controllers in the group.
2. **Database Lookup:** The old WLC responds with the client's details (MAC address, Security keys, IP).
3. **Context Transfer:** The user's session is transferred to the new WLC instantly. This happens so fast that Voice or Video calls do not drop.

3. Advantages of Cisco UWN Mobility

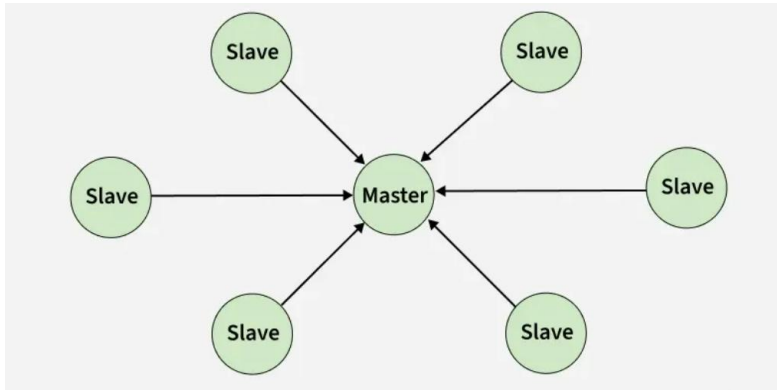
- **Seamless Roaming:** No disconnection during movement.
- **Scalability:** Supports thousands of APs across a large campus.
- **Security:** Keeps the user's security policies (VLAN, ACL) active regardless of location.

SIMPLE :- Cisco UWN Mobility ka matlab hai "**Roaming**". Jab tum college ke ek kone se dusre kone tak jaate ho, toh tumhara phone ek AP ko chod kar dusre AP se connect hota hai.

Agar dono APs ek hi subnet mein hain (Layer 2), toh connection easily shift ho jata hai. Par agar tum alag building (alag subnet) mein chale gaye, toh Cisco "**Anchor**" aur "**Foreign**" Controller ka logic use karta hai. Naya Controller purane wale se ek tunnel bana leta hai taaki tumhara IP address na badle aur tumhara call ya download chalta rahe. Isse "Zero Packet Loss" roaming milta hai.

Q13. Explain Bluetooth Architecture and Protocol Stack. (10 Marks)

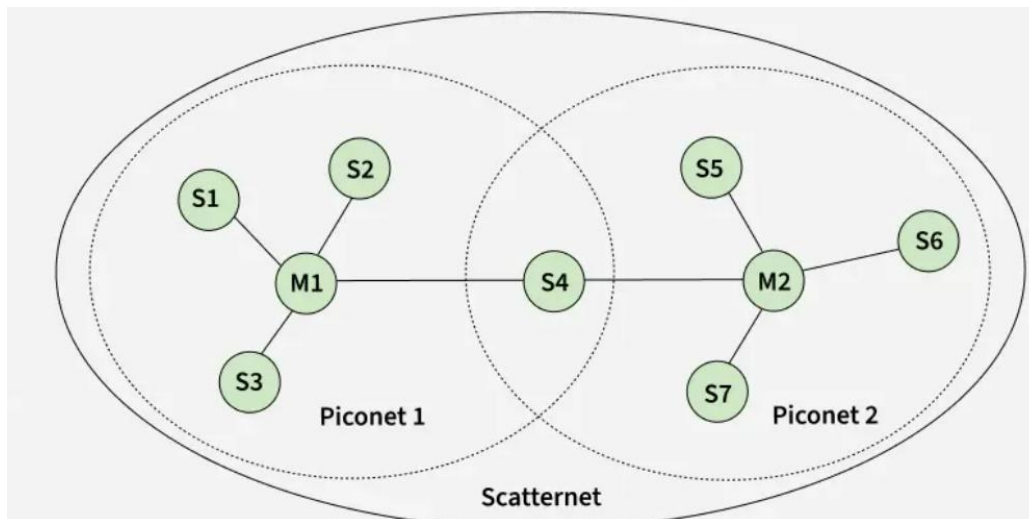
Step 1: Draw Piconet Architecture



Explanation:

- **Piconet:** It is the basic unit of Bluetooth.
- **Master:** Only **one Master** exists per piconet. It controls the timing and frequency hopping.
- **Slave:** Up to **7 active Slaves** can connect to one master. Slaves only transmit when allowed by the master.

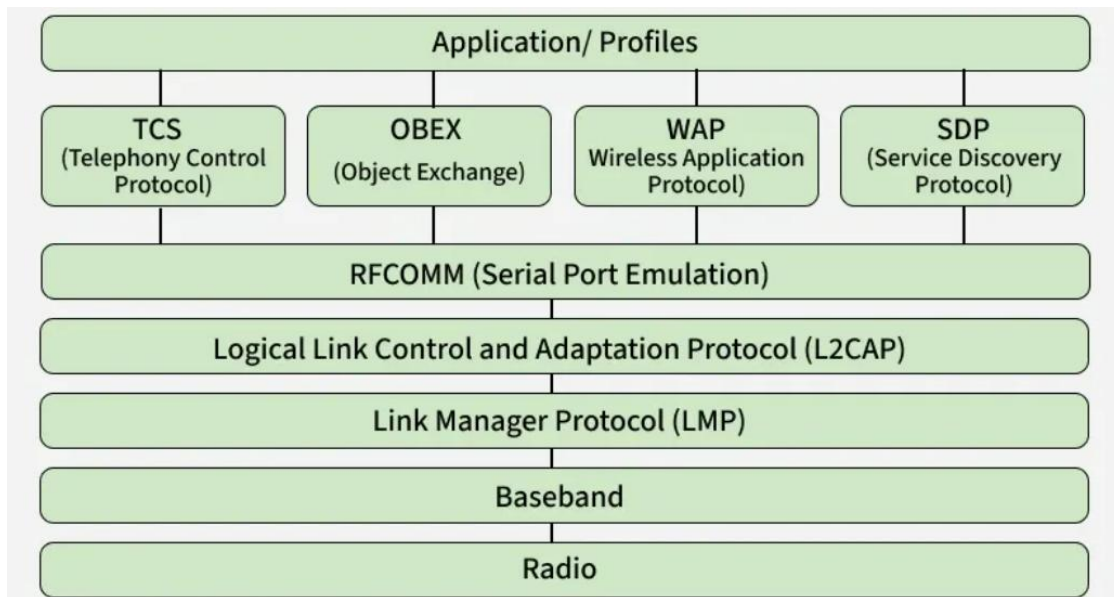
Step 2: Draw Scatternet Architecture



Explanation:

- **Scatternet:** It is formed by connecting two or more piconets.
- **Bridge Node:** A device (like **S4** in your diagram) acts as a bridge. It is a member of multiple piconets, allowing data to move between them.

Step 3: Draw Bluetooth Protocol Stack



Explanation of Layers :

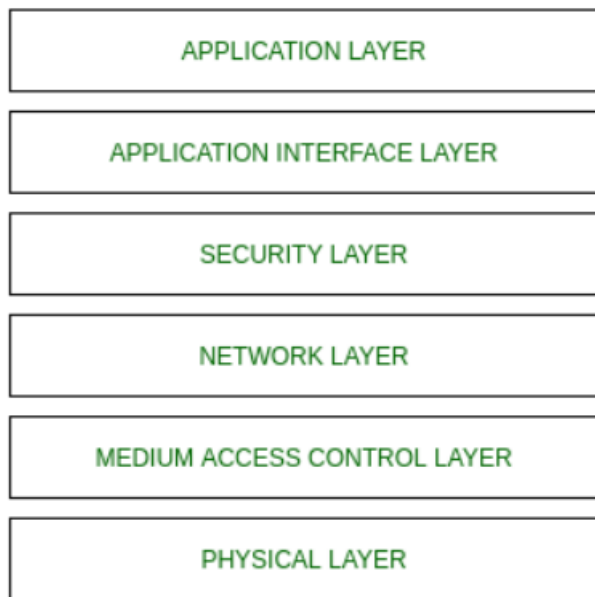
1. **Radio Layer:** Handles the physical 2.4 GHz radio waves.
2. **Baseband:** Manages the Master-Slave connections and timing.
3. **LMP (Link Manager Protocol):** Handles **Security**, pairing, and encryption.
4. **L2CAP:** Performs **Segmentation** (breaking big data into small packets) and Reassembly.
5. **RFCOMM:** Emulates a **Serial Port** to support old applications.
6. **SDP (Service Discovery):** Searches for available services on nearby devices.
7. **OBEX/TCS:** Used for **File Transfer** (Object Exchange) and Telephony calls.

Q14. What is Zigbee? Explain ZigBee protocol stack in detail.

Zigbee is a high-level communication protocol used to create Personal Area Networks (PAN). It is specially designed for low-power, low-data rate, and close-proximity wireless mesh networks.

- **Standard:** It is based on the IEEE 802.15.4 standard.
- **Frequency:** It operates in the 2.4 GHz ISM band.
- **Usage:** Mainly used in Home Automation, Smart Grids, and Industrial Control (IoT).

Zigbee Protocol Stack Architecture



1. Physical Layer (PHY):

- This is the lowest hardware layer.
- It handles the actual **radio signals** (2.4 GHz).
- It is responsible for turning data into radio waves and vice-versa.

2. Medium Access Control (MAC) Layer:

- It manages the connection between devices.
- It uses **CSMA/CA** (Carrier Sense Multiple Access) to make sure two devices don't send data at the same time and cause a collision.
- It handles **Beacons** for network synchronization.

3. Network Layer:

- This is the "Brain" of the network.
- It handles **Routing** (finding the best path for data).
- It allows new devices to join or leave the network.
- It supports Star, Tree, and **Mesh topologies**.

4. Security Layer:

- As shown in your diagram, Zigbee has a dedicated layer for safety.

- It provides **Encryption (AES-128)** to keep data private.
- It manages security keys to ensure only authorized devices can communicate.

5. Application Interface Layer:

- It acts as a bridge between the network and the actual user application.
- It handles **Binding**, which means connecting two devices based on their services (e.g., connecting a switch to a bulb).

6. Application Layer:

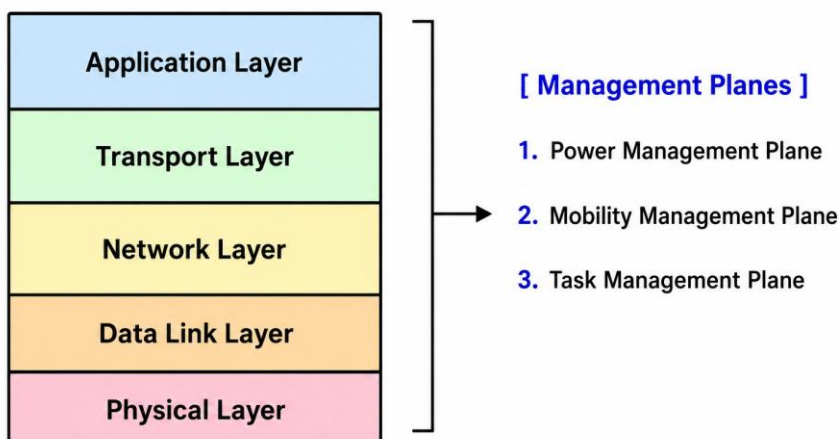
- This is the top layer where the actual work happens.
- It contains the software that controls the device (like a temperature sensor reading or a light-dimming command).

Q. Explain the Layered Architecture of WSN. (10 Marks)

Answer:

A Wireless Sensor Network (WSN) consists of many small sensor nodes that work together to monitor an environment. Its architecture is divided into 5 Layers and 3 Management Planes.

1. WSN Protocol Stack Diagram



2. Explanation of 5 Layers (Simple English)

1. Physical Layer:

- It handles the actual radio transmission and reception.
- It is responsible for frequency selection, signal encryption, and data rate.

2. Data Link Layer:

- It manages MAC (Medium Access Control).
- It minimizes collisions between neighbor nodes' signals and handles error detection.

3. Network Layer:

- Its main job is Routing. It finds the best path for data to travel from the sensor node to the Sink (Base Station).

4. Transport Layer:

- It ensures data reliability and helps in data flow control. It is important when the network needs to avoid congestion (jamming).

5. Application Layer:

- This layer makes the data useful for the user. Different types of software (like temperature monitoring or vibration analysis) run here.

3. The 3 Management Planes (Very Important)

These planes help sensor nodes work together efficiently to save battery:

- **Power Management Plane:** It manages how a sensor node uses its battery. For example, it turns off the radio when the node is not sending data to save power.
- **Mobility Management Plane:** It detects and keeps track of nodes that are moving.
- **Task Management Plane:** It balances the workload among nodes. It decides which node should sense and which should just pass data.

WSN architecture mein 5 layers toh normal networking jaisi hain, par 3 Planes sabse special hain:

1. **Power Plane:** Sensor ki battery bachane ke liye (Sabse important!).
2. **Mobility Plane:** Agar sensors move kar rahe hain toh unhe track karne ke liye.
3. **Task Plane:** Ye decide karne ke liye ki kis node ko kya kaam karna hai.

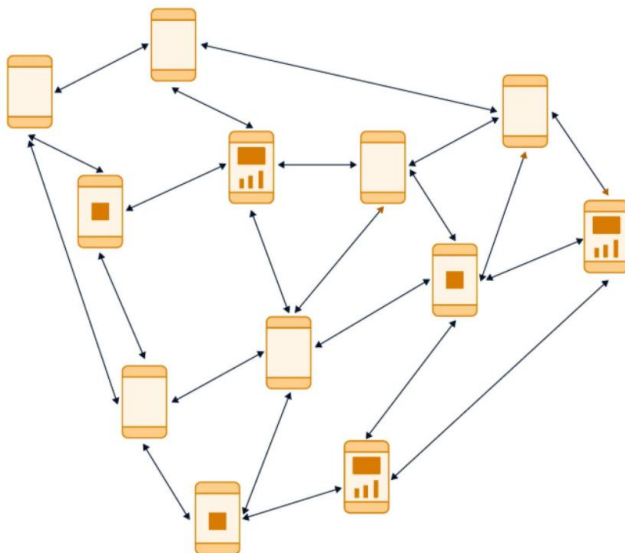
Theek hai Naved, MANET Architecture ka simple English mein detail answer ye raha. Exam mein tumhare diye huye diagram ke saath ye points likhna:

Q. Explain the Architecture of MANET (Mobile Ad-hoc Network). (10 Marks)

Answer:

MANET stands for Mobile Ad-hoc Network. It is a network of mobile devices that connect wirelessly without any central tower or fixed cables.

1. MANET Architecture Diagram



2. Key Components of the Architecture

- **Mobile Nodes:** These are the devices in the network, like smartphones, laptops, or drones.
- **Wireless Links:** The connections between these devices. Since nodes move, these links are temporary.
- **Infrastructure-less:** There is no fixed Base Station or Access Point.

3. Detailed Features

1. **Every Node is a Router:** This is the most important part. In a normal network, your phone only sends data. But in a MANET, your phone also acts as a Router to help pass someone else's data to the next device.
2. **Self-Configuring:** When a new device comes near the network, it automatically joins. You don't need a network engineer to set it up.
3. **Dynamic Topology:** "Topology" means the shape of the network. Because devices are always moving, the network shape changes every second.
4. **Multi-hop Routing:** If Node A is too far from Node C, it will send data to Node B first, which then passes it to Node C. This "jumping" is called multi-hop routing.
5. **Distributed Control:** There is no central "Boss" or Server. Every device has equal responsibility to manage the network.

Simple Explanation:

iska logic bahut simple hai. Socho ek aisi jagah jahan koi mobile tower nahi hai (jaise koi forest ya earthquake area). Wahan log apne phones ko aapas mein connect karke ek network bana lete hain.

1. Isme har mobile phone ek Router ki tarah kaam karta hai.
2. Agar tumhara friend tumse door hai, toh beech wala banda tumhara message aage pass karega.
3. Exam mein diagram ke saath "Self-Configuring" aur "Dynamic Topology" words zaroor likhna, marks ache milenge.